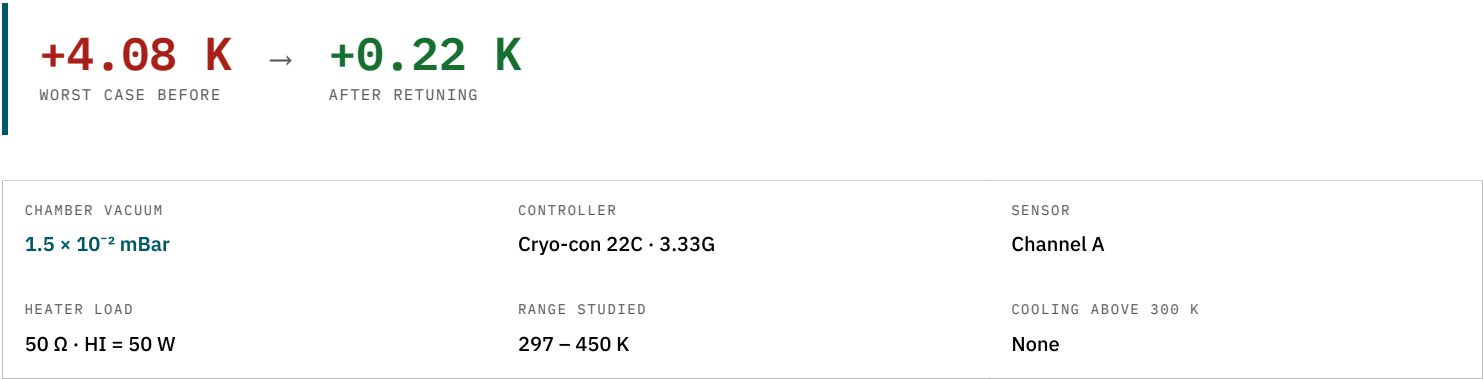


Eliminating temperature overshoot on the probe station

Why the station used to sail 4 K past its setpoint and sit there for half an hour — and the controller settings that fixed it.



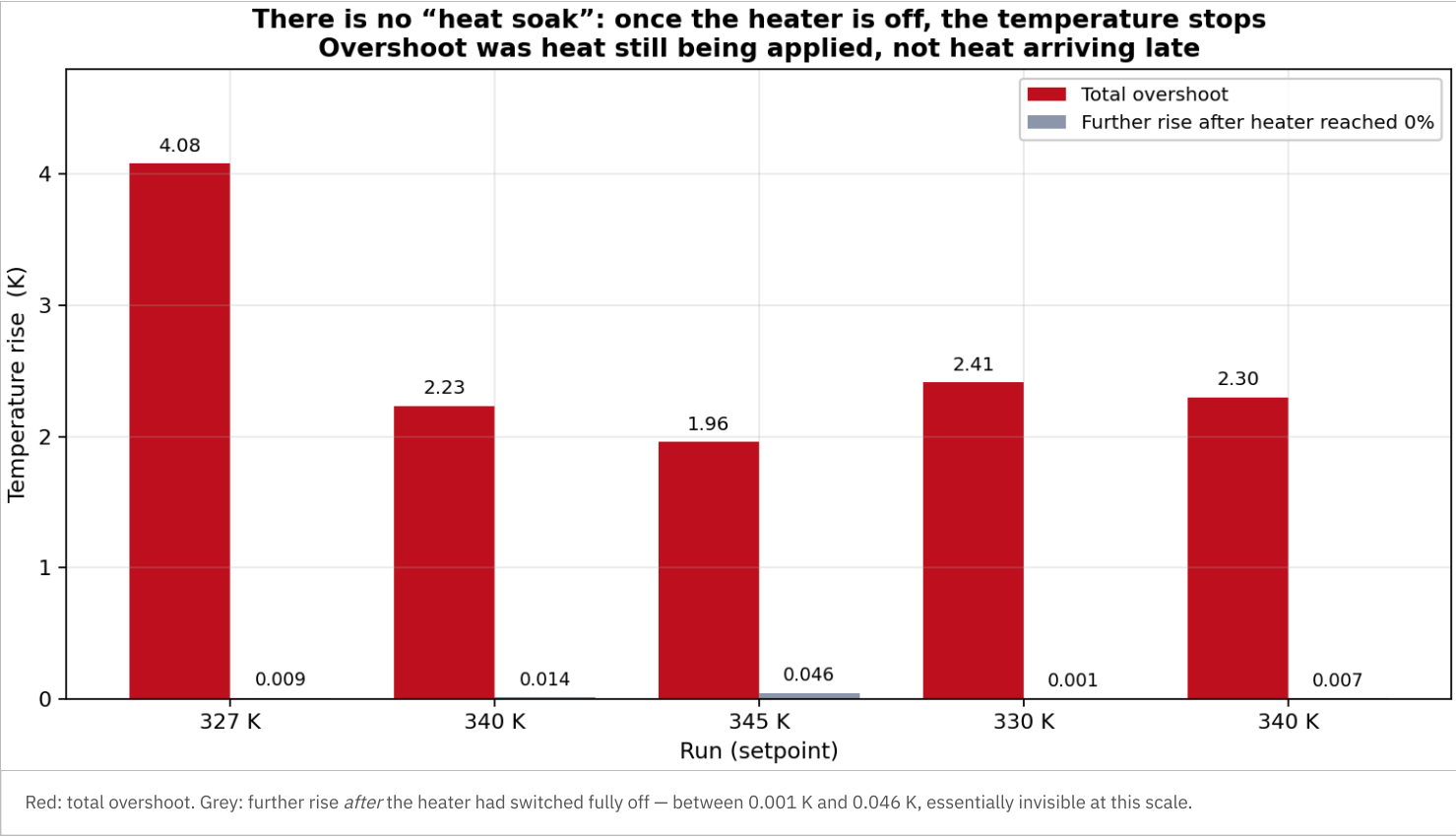
01 What was happening

Setting a new temperature sent the station 2 to 4 degrees past it. Above 300 K there is no active cooling — only the heater — so the only way back down is to lose heat naturally, at about a quarter of a degree per minute. A 3 K overshoot stranded the experiment for half an hour.

Two explanations had been written into the project documentation. Both turned out to be wrong, and one of them had been sending the tuning in the wrong direction for months.

02 The "heat soak" theory was wrong

The documentation explained overshoot as heat continuing to soak into the sample after the setpoint is reached. That is testable: find the moment heater output actually reaches zero, then measure how much further the temperature rises.



The temperature stops climbing almost the instant the power stops. **There is no stored-heat lag.** In every run the overshoot happened because the heater was *still switched on, at 27–29 % power, while the sample was already above setpoint* — for a consistent 166 to 170 seconds each time.

03 One setting had the wrong units

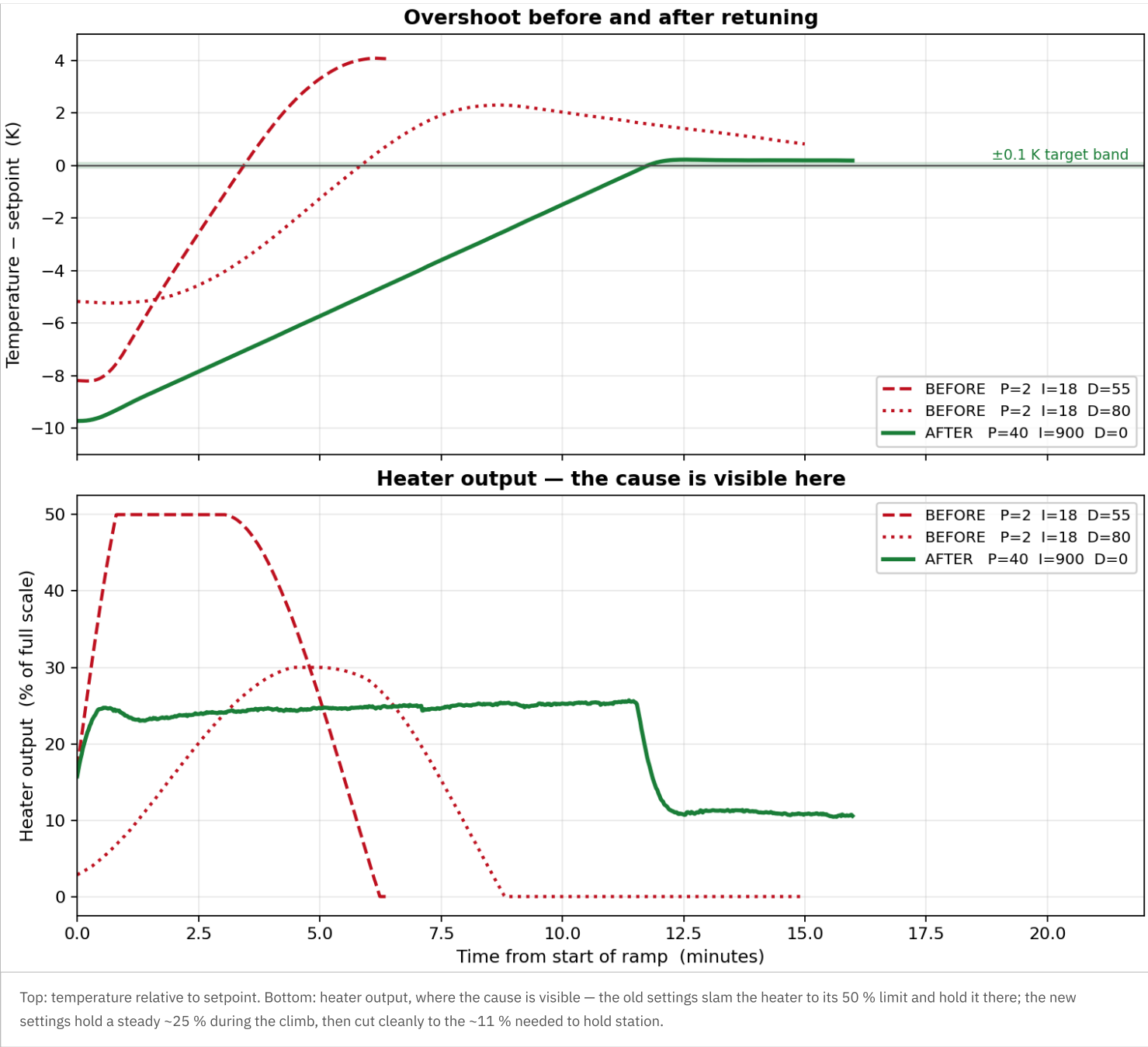
On this controller `IGAIN` is not a gain. It is the **integral time, in seconds** — the manual states plainly that it is "a numeric field with units of seconds". A *smaller* number makes the integrator *faster and more aggressive*, not gentler.

Every previous attempt lowered it to fight overshoot — 160, then 30.4, then 18 — making the problem worse each time. The project's own table-writing script still carried the comment "*I = 30–45 → ~5× lower than original. Prevents integrator windup.*" That reasoning is exactly inverted. Standard practice for thermal systems is 60–300 seconds or more; the final value is 900.

ALSO BLOCKING THE TARGET

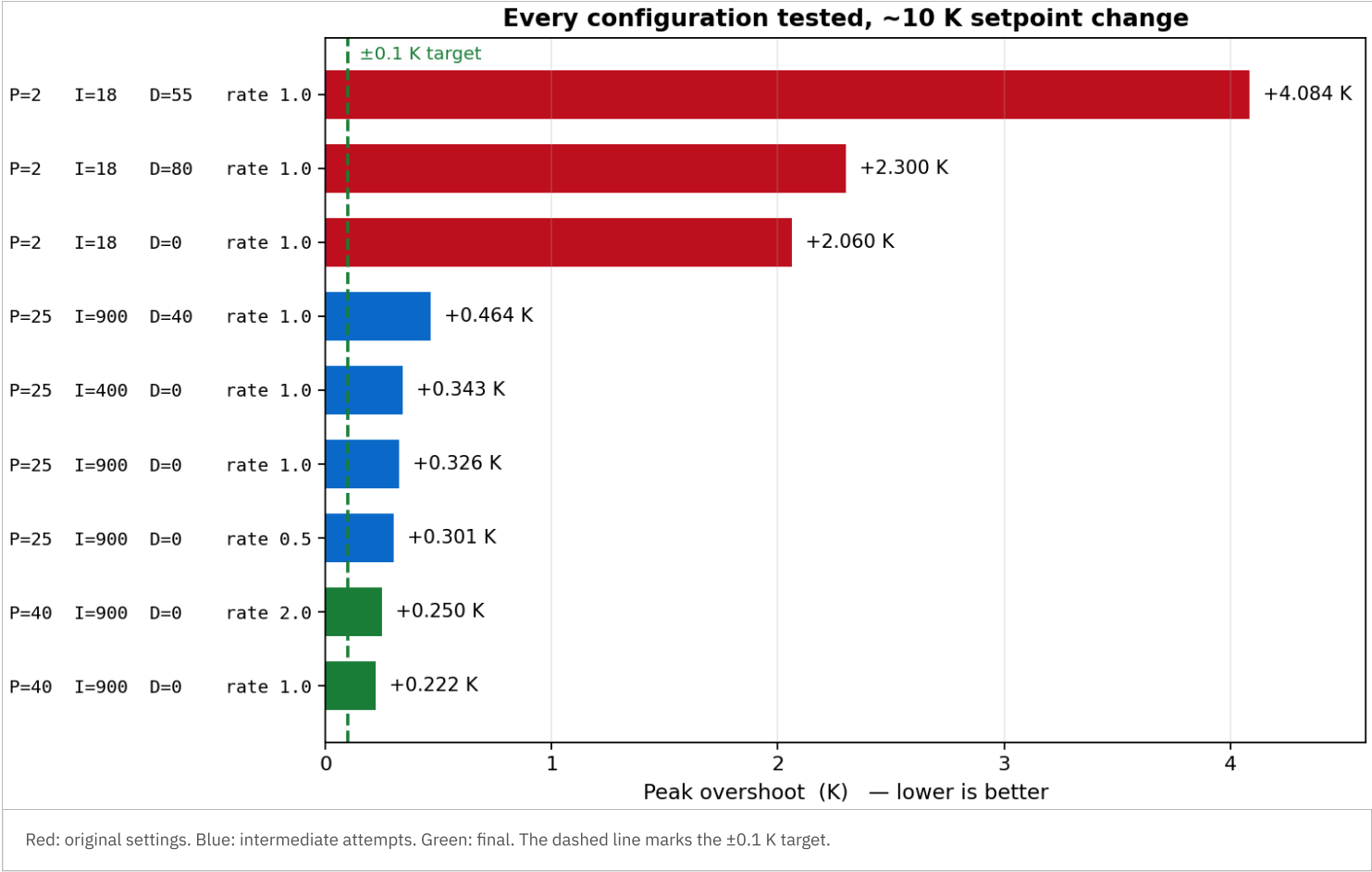
An earlier session capped heater power at 30 %. Holding 450 K needs about 21 %, and *ramping* at 1 K/min at 450 K needs about 39 %. The cap made the target physically unreachable — and guaranteed the heater ran pinned at its limit, which is what wound the integrator up in the first place.

04 Before and after



05 Everything that was tested

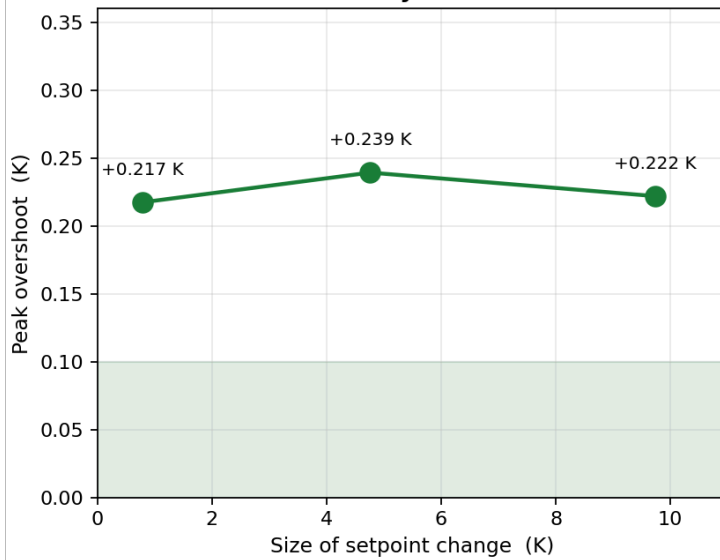
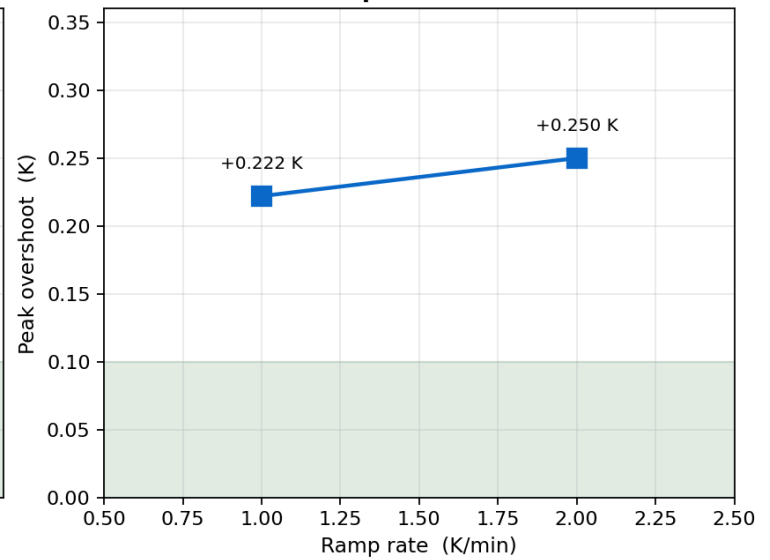
Nine hardware runs, each changing one thing at a ~10 K setpoint change.



CHANGE	RESULT	VERDICT
Integral time 900 → 400 s	+0.222 → +0.343 K	Worse
Derivative 0 → 40	+0.326 → +0.464 K	Worse
Ramp rate 1.0 → 0.5 K/min	+0.326 → +0.301 K	No real change
Ramp rate 1.0 → 2.0 K/min	+0.222 → +0.250 K	No real change
Proportional gain 25 → 40	+0.326 → +0.222 K	The only thing that helped

06 Overshoot doesn't depend on how far you move

This was the most useful finding of the study, and it came from watching the live traces rather than from theory.

Both plots at $P=40$, $I=900$, $D=0$ — only P changes the outcome**Overshoot is independent of how far you travel****...and barely depends on ramp rate either**

Left: setpoint changes of 0.8 K, 4.8 K and 9.7 K all produced the same ~0.22 K overshoot. Right: doubling the ramp rate barely moved it either.

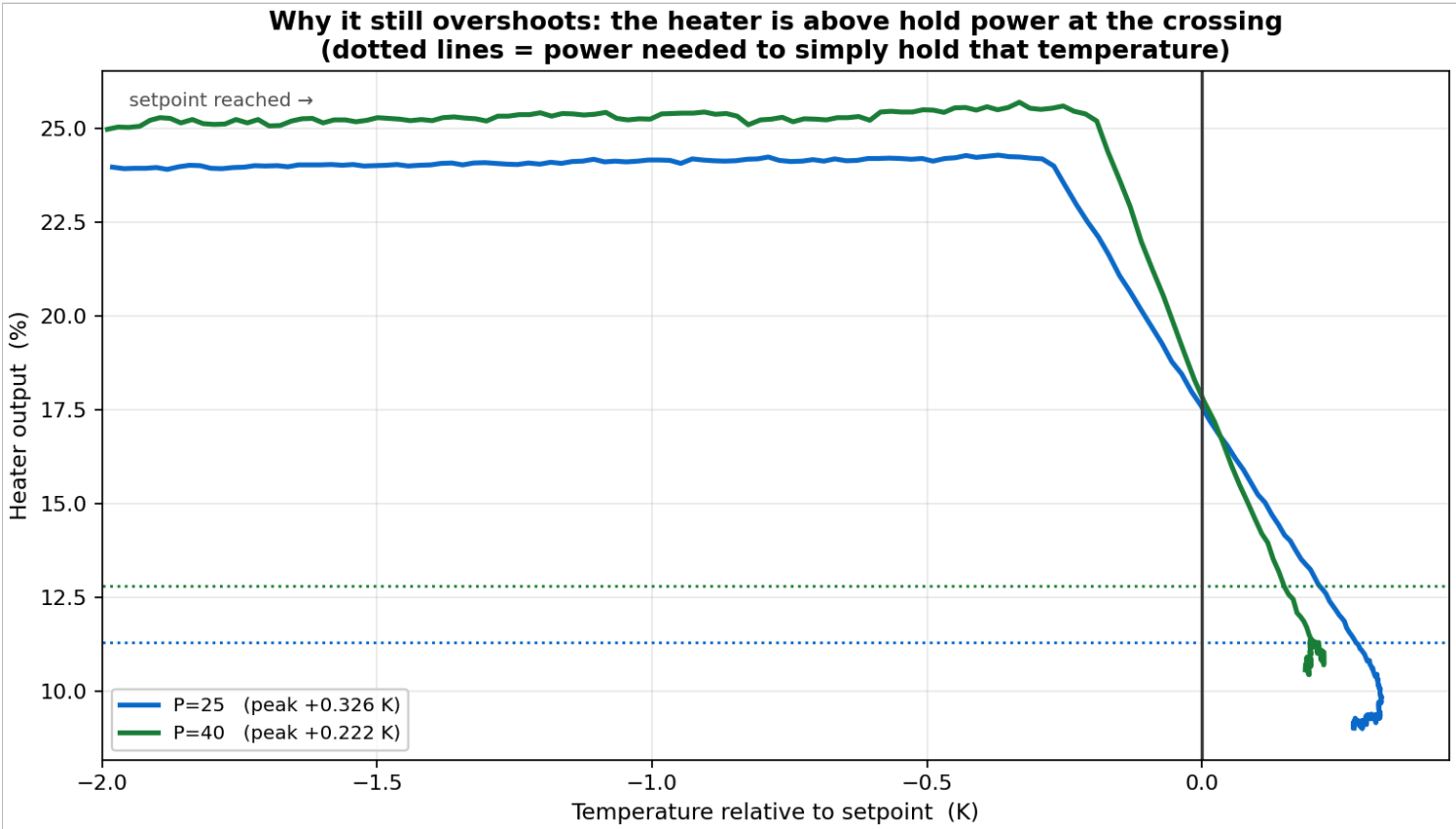
Why: the surplus heat that has to be shed on arrival is the *ramp power* — the extra wattage needed to make the temperature climb at all. That depends only on how fast you are climbing, not how far you have climbed. Move 0.8 K or 10 K, you arrive carrying the same surplus, so you overshoot the same amount.

This single fact explains every other result: why hop size is irrelevant, why integral time and derivative are irrelevant (neither changes the ramp power), and why only proportional gain helps — it is the only term that converts the tracking lag into shed power.

PRACTICAL CONSEQUENCE

Breaking a large temperature change into small steps does **not** help. Go directly to the temperature you want.

07 Where the floor is, and why



Both traces sit flat during the climb, then turn over sharply — but are still above the dotted "hold power" line when they cross zero. That surplus is the overshoot. P=40 turns over later yet falls far more steeply.

The heater only begins backing off when the controller's internal moving setpoint finishes its ramp. Before that the error is constant, so the output is constant — the controller has no way of knowing the ramp is about to end.

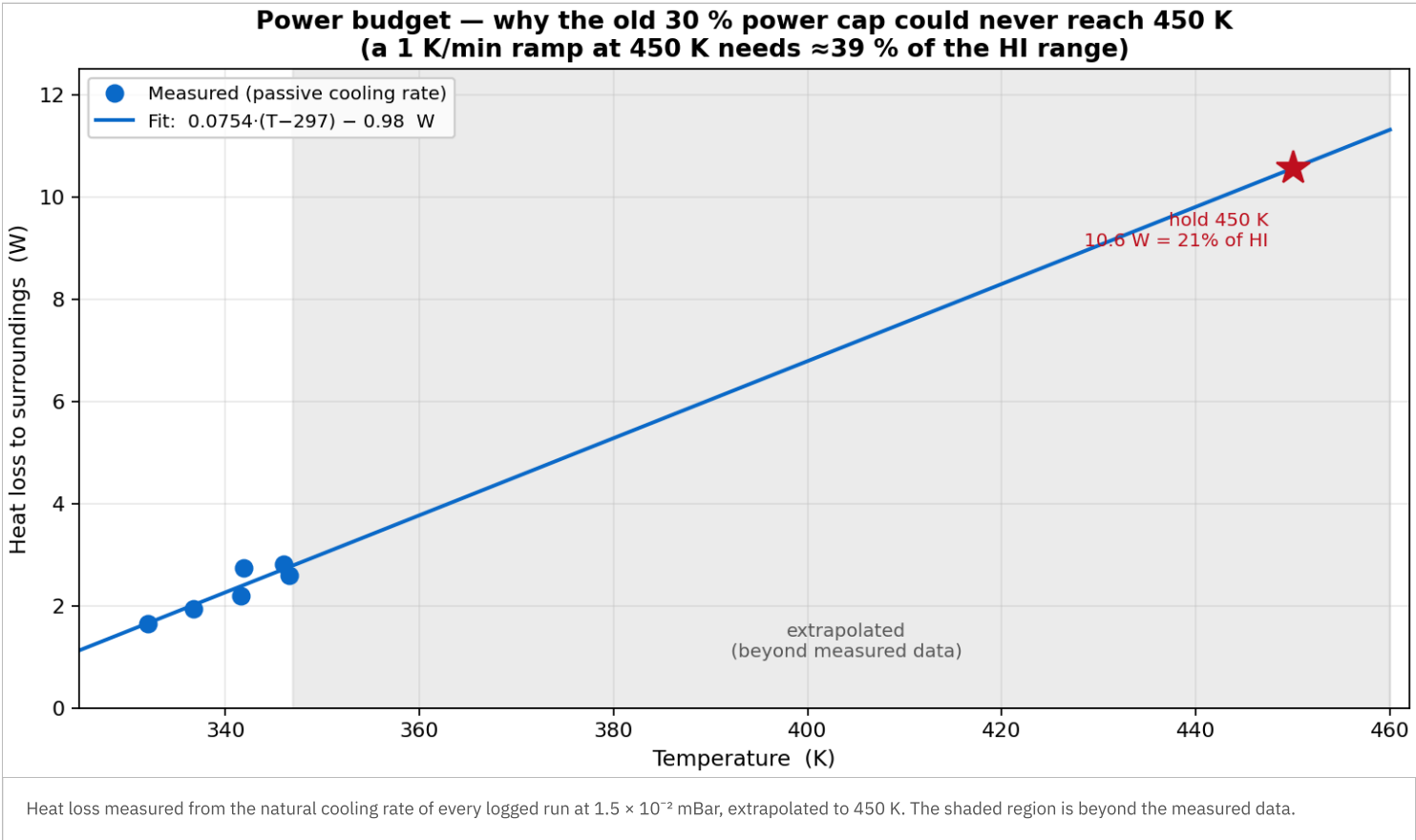
```
surplus to shed = ramp power 27.3 % - hold power 15.1 % = 12.2 %
shed by P term  = P 40 × tracking lag 0.16 K           = 6.4 %
predicted surplus at crossing                          = 5.8 %
MEASURED surplus at crossing = 20.8 % - 15.1 %         = 5.7 % ✓
```

Driving that remaining surplus to zero would need **P ≈ 76**. The estimated stability limit for this loop is **P ≈ 82** — so zero overshoot sits essentially on the stability boundary with no margin, and around P ≈ 70 the loop would be expected to start ringing.

HONEST LIMIT

≈0.2 K is the floor for this controller using a single flat ramp. Reaching ±0.1 K at the moment of arrival would require reducing the rate during the final approach — which means a script changing the rate mid-ramp, the behaviour ruled out for the LabVIEW deployment.

08 Power budget to 450 K



TEMPERATURE	POWER TO HOLD	HOLD + RAMP AT 1 K/MIN
350 K	3.0 W (6 %)	12.2 W (24 %)
400 K	6.8 W (14 %)	15.9 W (32 %)
450 K	10.6 W (21 %)	19.7 W (39 %)

450 K was reached and held during the overnight campaign, with measured holding power of **20.3–20.9 %** against the 21.1 % predicted here. Note these figures are specific to 1.5×10^{-2} mBar — a materially different vacuum changes the gas conduction and therefore the power needed, though not the tuning conclusions.

09 Final settings

LOOP 1:PGAIN	40	proportional gain
LOOP 1:IGAIN	900	integral time, SECONDS
LOOP 1:DGAIN	0	derivative off
LOOP 1:RANGE	HI	50 W full scale
LOOP 1:MAXPWR	70	percent
LOOP 1:RATE	1.0	K/min – 2.0 is equally good and twice as fast
LOOP 1:TYPE	RAMPP	ramp using these gains, NOT a table lookup

DO NOT USE RAMPT OR TABLE MODE

In those modes the controller **ignores the gains you send** and reads them from PID Table 2 instead — which still contains the old bad values (P≈2, I=30–45, D=52–70).
LOOP 1:TABLEIX? confirms Loop 1 is still pointed at that table.

09b PID Table 2 as written to the instrument

Written to the controller's non-volatile memory on **2026-09-05** and verified by reading all 16 rows back on a fresh connection — 16 of 16 correct. The previous contents were saved to pid_table02_BACKUP_20260905_092227.txt first.

This matters because LOOP 1:TABLEIX? returns 2 — Loop 1 is bound to this table. Until now, any switch to RAMPT or TABLE mode would have silently restored the old, bad gains.

SETPOINT (K)	WAS — P / I / D	NOW — P / I / D	RANGE
ZONE 1 — 300 TO 475 K · NO ACTIVE COOLING · RETUNED			
475	3.00 / 45.00 / 70.00	40.00 / 900.00 / 0.00	HI
450	2.90 / 43.00 / 68.00	40.00 / 900.00 / 0.00	HI
425	2.70 / 40.00 / 65.00	40.00 / 900.00 / 0.00	HI
400	2.60 / 38.00 / 63.00	40.00 / 900.00 / 0.00	HI
375	2.40 / 36.00 / 60.00	40.00 / 900.00 / 0.00	HI
350	2.30 / 34.00 / 58.00	40.00 / 900.00 / 0.00	HI
325	2.10 / 32.00 / 55.00	40.00 / 900.00 / 0.00	HI
300	2.00 / 30.00 / 52.00	40.00 / 900.00 / 0.00	HI
ZONE 2 — 77 TO 275 K · LN ₂ COOLING · LEFT UNTOUCHED, NO DATA YET			
275	unchanged	2.00 / 55.00 / 50.00	HI
250	unchanged	1.90 / 58.00 / 47.00	HI
225	unchanged	1.80 / 60.00 / 44.00	HI
200	unchanged	1.70 / 62.00 / 41.00	HI
175	unchanged	1.60 / 62.00 / 38.00	HI
150	unchanged	1.50 / 60.00 / 33.00	HI
120	unchanged	1.30 / 56.00 / 27.00	HI
77	unchanged	1.00 / 50.00 / 20.00	MID

DISCONTINUITY AT THE ZONE BOUNDARY

The table interpolates between rows, so between 275 K and 300 K the gains swing steeply from P=2 to P=40. That region is only reachable with LN₂ cooling active, which the station does not currently have — but it should be retuned before the first cryogenic run rather than trusted as-is.

PERSISTENCE

The table is held in non-volatile memory and survives a normal front-panel power-down. Per the manual, SYSTEM:NVSAVE additionally commits the entire instrument configuration to flash so it is restored after an *abrupt* power loss — worth running if the station's mains is ever cut without using the front panel switch. It was **not** run here, since it would also freeze every other current setting as the power-up default.

10 LabVIEW instructions

Serial: **COM5, 9600 baud, 8 data bits, no parity, 1 stop bit**. Terminate every command with carriage-return + line-feed (`\r\n`). Allow roughly **200 ms** between commands — the controller drops commands sent faster than it can process them. Queries end with `?` and return one line.

1 Once at the start of a session

```

LOOP 1:RANGE HI
LOOP 1:MAXPWR 70
LOOP 1:PGAIN 40
LOOP 1:IGAIN 900
LOOP 1:DGAIN 0
LOOP 1:RATE 1.0

```

Verify with the matching query if you want confirmation — LOOP 1:PGAIN? should return 40.000000 .

2 For every temperature change

INPUT? A	read current temperature → T _{now}
LOOP 1:TYPE PID	leave ramp mode so the next line is instant
LOOP 1:SETPT <T _{now} >	park working setpoint at where you are
LOOP 1:TYPE RAMPP	back into ramp mode
CONTROL	loop ON — must precede the target
LOOP 1:SETPT <target>	LAST — this is what starts the ramp

WHY THAT ORDER MATTERS

CONTROL must come before the target setpoint. The ramp only runs while the loop is controlling. Send the target while control is OFF and no ramp is armed — then the moment control switches on, the controller sees the entire error at once. With P=40 a 10 K error demands 400 % power, so the heater slams to its limit. Observed directly: heater pinned at 70 % for 35 seconds, winding the integrator to 60 % when 7 % was needed.

The setpoint must be last, because writing it is what arms the ramp. Anything changed afterwards is not applied to the ramp in progress.

On the current front panel this means pressing **Start Control before Set Temperature** — the panel's default order causes the surge above.

3 Monitoring while it runs

INPUT? A	temperature, K
LOOP 1:OUTP?	heater output, % of full scale
LOOP 1:RAMP?	YES while ramping, NO when finished
CONTROL?	ON or OFF

4 Shutting down

STOP	disengage loop, heater to 0
CONTROL?	confirm it returns OFF

Always confirm CONTROL? returns OFF . If a program is killed without sending STOP , the controller keeps heating to its last setpoint indefinitely.

Front panel items that need fixing

CONTROL	ISSUE
P / I / D defaults	Shows 1.00 / 5.00 / 0.00 — near factory defaults, far worse than tuned. Set 40 / 900 / 0.
Ramp Rate default	Shows 0.01 K/min. At that rate 273 → 450 K takes about 12 days. Set 1.0.
Range default	Shows LOW (0.5 W). Cannot heat on LOW or MID above ~330 K. Set HI.
LOOP 1: TYPE	No control exists. Without it you cannot guarantee RAMPP, and in Table mode your PID values are silently ignored. Add it.
LOOP 1: MAXPWR	No control exists. Add it or hard-code 70. Never 30.
"Set Resistor Value"	Labelled for resistance but wired to a Range dropdown. Confirm it sends LOOP 1: RANGE , not LOOP 1: LOAD — load must stay 50 Ω or every p

11 What to expect in practice

MEASURE	BEFORE	AFTER
Peak overshoot	+2.3 to +4.1 K	+0.22 K
Depends on jump size?	—	No
Depends on ramp rate?	—	Barely
After arrival	stranded 30 min+	smooth decay, then steady
450 K reachable?	No — power cap too low	Yes, verified

Set the temperature, let it arrive, allow a short settling period, then measure. Once inside the ±0.1 K band it stays there indefinitely — stability was never the problem, only the approach.

12 Limitations

1. **±0.1 K is not achieved at the instant of arrival.** It needs a short settling wait. Closing that gap requires reducing the ramp rate during the final approach.
2. **PID Table 2 has now been rewritten** with the tuned Zone 1 values and verified (§09b). Zone 2 is untouched, so the 275–300 K interpolation region is not yet trustworthy.
3. **The 77–300 K cryogenic zone was never examined.** Its settings are likely wrong for the same units reason, but with LN₂ cooling active the dynamics differ and there is no data. *LN₂ is on order* — retune this zone once it arrives, using the same single-variable method rather than assuming the Zone 1 values transfer.
4. **The heat-loss model is extrapolated** above ~347 K. The 450 K prediction matched well, but it remains an extrapolation and is specific to 1.5 × 10⁻² mBar.
5. **The P ≈ 82 stability limit is calculated, not measured** — testing it means deliberately driving the loop into oscillation.

13 Safety configuration

OVER TEMPERATURE DISCONNECT

Enabled, Channel A, 460 K — set by hand on the front panel. It is *not* remotely settable; every SCPI variant returns NAK , so no script can set, read back, or restore it. **If the instrument is ever reset it must be re-entered manually**, and nothing will warn you it is missing.

- Channel B reads a sensor fault, so Channel A is the only usable OTD source.
- The OTD is the only hardware-level protection — it opens a mechanical relay, so it works even if the fault is inside the controller's own output circuitry.
- Use staged_ramp_test.py for unattended runs; it sends STOP on every exit path. pid_tune_test.py does not, and will leave the heater running if the process is killed.

24 logged hardware runs · raw CSVs and backup in the project repository
· figures regenerated by `make_final_figures.py` · full technical log
in `OVERSHOOT_TUNING_LOG.md`